GZero 1.3 and 1.4: new analysis methods

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GZero 1.3 adds three methods that are standard in break-junction papers: the junction yield with per-trace plateau statistics, single-level model fits of I-V curves, and transition voltage spectroscopy. GZero 1.4 adds the analysis methods of Rashid et al., JACS 2025 (finite-temperature energy-grid fit, noise protocol, event detection, dwell time) and an EGaIn ensemble tab built after the Hong (XMe) and Chiechi (GaussFit) codes.

What's new

Each new method answers a question the existing tabs couldn't. All of them run on data GZero already loads, so no new file formats are needed.

Addition	Where	Question it answers	Module
Junction statistics	New tab after Plateau length	How many traces actually had a molecule, at what conductance, and did that drift during the run?	`gzero/junctions.py`
Single-level model fit	I-V tab	Where is the molecular level (ε_0) and how strongly is it coupled to the electrodes (Γ)?	`gzero/ivmodels.py`
Transition voltage spectroscopy	I-V tab	Where does the Fowler-Nordheim plot turn over, and what level position does that imply?	`gzero/ivmodels.py`
Events	New tab	Which traces flicker or switch mechanically?	`gzero/events.py`
EGaIn (ensemble)	New tab	Statistics of large-area junctions, yield, rectification and β	`gzero/egain.py`

One existing function got a fix along the way. The averaged I-V curve used to take the overlap of all sweeps, so the half sweeps at the start and end of a recording could shrink it to nothing. It now averages only sweeps that cover the typical bias range.

Papers reviewed

I went through the methods sections of commonly cited break-junction papers and looked for analyses people report routinely that GZero didn't have yet. Four papers define what was added in 1.3:

Paper	What they do	What GZero does with it
Kamenetska et al., PRL 102, 126803 (2009)	Count how many breaking traces show a molecular plateau (junction formation probability) and how long the plateaus are, for a series of molecules	Junction yield per trace with a binomial error, plus yield and conductance in blocks of traces over the run

Paper	What they do	What GZero does with it
Zotti et al., Small 6, 1529 (2010)	Fit MCBJ I-V curves of tolanes with thiol, nitro and cyano anchors with a single-level model to get ε_0 and Γ	Single-level fit of every curve and of the mean forward and backward curves, with errors and an optional bias asymmetry
Beebe et al., PRL 97, 026801 (2006)	Plot ln(I/V^2) against 1/V and read off the minimum, the transition voltage V_t; it scales with the HOMO offset from photoemission	V_t for both polarities of every curve, with a Fowler-Nordheim plot of the mean curves
Baldea, PRB 85, 035442 (2012)	Shows that both polarities of V_t together give the level position ε_0 and the bias asymmetry in closed form	ε_0 and the asymmetry a from V_t^+ and V_t^-, so TVS and the fit can be compared curve by curve

The 2D correlation, clustering and flicker-noise methods from earlier papers were already in GZero. I left out methods that need hardware most setups don't have, like thermopower and shot noise.

Junction statistics

A trace counts as a molecular junction if it stays flat inside the molecular window for at least 0.1 nm. The yield is the fraction of traces that do.

The tricky part is "flat". A trace that only tunnels still passes through a two-decade window in about 0.1-0.2 nm, so a plain length-in-window test counts almost every trace. GZero therefore checks the local slope at each point:

$$s(z) = \frac{|\log G(z + w/2) - \log G(z - w/2)|}{w}$$

w is 0.05 nm by default. Tunnelling falls at β / ln 10 decades per nm: about 9 for gold in vacuum and 4-5 in solvents. Molecular plateaus usually fall by well under 1 decade per nm. A point counts as flat below 2 decades/nm, which sits between the two.

Per trace you get three numbers:

- **flat length**: summed distance of the flat points inside the window
- **plateau conductance**: median log G of those points
- **junction yes/no**: flat length of at least *min plateau*

From these the tab gives the yield p with its binomial error sqrt(p(1-p)/N), the median plateau conductance and length, and a drift in decades per 1000 traces. The drift is a straight line fitted to the block medians, with 100 traces per block by default. The five plots are the plateau conductance histogram, conductance against trace number, yield against trace number, the flat-length histogram with the threshold marked, and length against conductance.

Use peak from histogram fit sets the window to the fitted peak +/- 3 σ. Then look at the flat-length histogram. Tunnelling traces pile up near zero and molecular ones form a second hump, and the threshold belongs in the dip between them.

Single-level model

The fit describes each I-V curve with one molecular level at energy ε_0 from the Fermi level, broadened by the coupling Γ to each electrode. At zero temperature the Landauer current through a Lorentzian transmission has a closed form:

$$I(V) = NG_0\Gamma\left[\arctan\frac{V/2-\varepsilon(V)}{\Gamma}+\arctan\frac{V/2+\varepsilon(V)}{\Gamma}\right], \quad \varepsilon(V)=\varepsilon_0+aV$$

Energies are in eV and V in volts, so G_0 times an energy in eV is directly a current in amperes. Both electrodes are coupled equally. N is the number of molecules in parallel and stays at 1. a lets the level move with the bias in an asymmetric junction; it is 0 unless you tick *fit asymmetry a*. At low bias the model gives $G/G_0 = \Gamma^2 \,/\, (\varepsilon_0^2 + \Gamma^2)$.

How the fit works:

1. The current is scaled by its largest value so all curves weigh the same.
2. A least-squares fit starts from four guesses of ε_0 (0.4, 0.8, 1.5 and 3 times the largest bias). Each guess gets a Γ that matches the measured low-bias conductance.
3. The fit with the smallest residual is kept. Errors come from the fit covariance, and R^2 is reported.
4. This is done for every curve (up to 500), for the mean forward and backward curves, and for the most probable curve.

The model can't tell HOMO from LUMO transport, because both give the same I-V. ε_0 is therefore reported as a positive number.

Transition voltage spectroscopy

The transition voltage V_t is where $V^2/|I|$ peaks, which is the same as the minimum of the Fowler-Nordheim plot, $\ln(|I|/V^2)$ against $1/V$. It needs no fit, which is why so many papers report it.

For each curve and polarity GZero averages the current in 60 bias bins above 0.05 V and finds the largest $V^2/|I|$. It then refines that point with a parabola through it and its two neighbours. If the maximum sits at the end of the sweep, the bias never reached V_t, and the value is left empty rather than guessed.

In the single-level model with Γ much smaller than ε_0, the two polarities give the level position and the asymmetry exactly (Baldea 2012):

$$\varepsilon_0 = \frac{2|V_t^+ V_t^-|}{\sqrt{(V_t^+)^2+(V_t^-)^2+\frac{10}{3}|V_t^+V_t^-|}}, \quad a = \frac{(V_t^+ + V_t^-)\,\varepsilon_0}{4|V_t^+V_t^-|}$$

For a symmetric junction this reduces to $\varepsilon_0 = (\mathrm{sqrt}(3)/2)\ V_t$, about 0.87 V_t. The a formula follows from the same stationarity condition and was checked numerically against the exact model (next section). Comparing the ε_0 from TVS with the ε_0 from the fit is a quick test of whether a single level describes the curves at all.

Validation of the 1.3 methods

On data with known answers, every new method recovers the true value within its error. The SLM and TVS rows are model curves with 2 % noise. The yield rows are GZero's own simulator going through the normal trace detection.

Test	True value	GZero
SLM fit, symmetric	ε_0 0.800 eV, Γ 10 meV	0.801 eV, 10.0 meV (R^2 0.9997)
SLM fit, asymmetric	ε_0 0.800 eV, a 0.100	0.795 eV, a 0.096
TVS, symmetric	ε_0 0.800 eV	V_t = +/-0.924 V, ε_0 0.800 eV

Test	True value	GZero
TVS, asymmetric	ε_0 0.800 eV, a 0.100	V_t^+ 1.191 V, V_t^- -0.746 V, ε_0 0.800 eV, a 0.100
TVS, asymmetric	ε_0 1.000 eV, a -0.080	ε_0 1.002 eV, a -0.080
Self-test I-V	ε_0 0.800 eV, Γ 20 meV	fit 0.800 eV, 20.0 meV; TVS 0.801 eV
Junction yield, 200 traces	70 %	68.5 +/- 3.3 %
Junction yield, 200 traces	30 %	29.0 +/- 3.2 %
Plateau conductance	log G -3.5	median -3.55

The slight shift of the plateau conductance comes from the simulated plateau slope (0.3 decades/nm), which pulls the median a little below the plateau's start.

Version 1.4: methods from Rashid et al. 2025

Every analysis section of the SI of Rashid et al., JACS 147, 830 (2025) is now in GZero, except the DFT and transport calculations.

SI section	What the paper does	In GZero
2.3 Flicker noise	Hold 160 ms at 100 mV, drop 10 ms, keep holds whose first and last 100 points sit within +/- 1 σ of the peak, DFT squared, 100 Hz-1 kHz, n where Pearson r is smallest (0.3-2.3, step 0.01)	Flicker noise tab, preset "Rashid et al. 2025"; new *end points* setting; minimum-|r| grid value reported
2.4 I-V fit	Landauer integral with Fermi functions on an energy grid (-10 to 10 eV, 0.02 meV), $\Gamma_L = \Gamma_R$, Levenberg-Marquardt fit of the most probable I-V	I-V tab, preset "Rashid et al. 2025"; energy grid or exact digamma form; most probable curve from a Gaussian per bias bin
2.5 Mechanical modulation	2D conductance-time histograms of holds with the most probable G per time	Events tab with segments = holds
2.6 Event detection	Butterworth 200 Hz, Savitzky-Golay (order 20, 20 side points), derivative, peak thresholds 0.0007 / 0.002, width 50	Events tab
2.7 Dwell time	τ = A/w x 1000/f_s ms from the peak area	Histograms tab, with every peak fit

At finite temperature the Landauer integral is evaluated either on the energy grid of the paper or with the exact closed form, which uses the digamma function psi:

$$\int T(E)\, f(E-\mu)\, dE = \pi\Gamma \left[\frac{1}{2} - \frac{1}{\pi}\,\mathrm{Im}\,\psi\!\left(\frac{1}{2} + \frac{\Gamma - i(\mu-\varepsilon)}{2\pi k_B T}\right)\right]$$

The two agree to 10^{-9}. Fits therefore start from the closed form and the grid only polishes, which brings a grid fit down from 50 s to under 1 s.

Test	True	GZero 1.4
Preset fit, model curve with the G1 values of Table S1, 300 K	ε_0 1.10 eV, Γ 11.3 meV	1.100 eV, 11.30 meV
Same, 2 % noise, through the GUI (mean, most probable, single curves)	1.10 eV, 11.3 meV	1.098-1.100 eV, 11.3 meV
Noise preset, 5 simulated sets, holds on the plateau	n = 1.5	1.53 (Morris preset: 1.59)

Test	True	GZero 1.4
Dwell time, 70 % molecular, 0.4 nm at 2 nm/s	140 ms	135-146 ms

The SI gives no temperature, so the preset assumes 300 K. Change it if your measurements ran at another temperature.

Events tab

The tab finds flickering and mechanical switching in G(t) with the four steps of SI 2.6, and classifies each trace as mechanical, flickering or quiet.

1. Butterworth low-pass, 200 Hz, order 2, applied forwards and backwards so events aren't shifted in time.
2. Savitzky-Golay smoothing, order 20 over 41 points. scipy's own coefficients collapse to about 10^{-17} at this order, so GZero builds the filter from a Legendre-basis QR. It matches scipy to 10^{-14} at low order.
3. Second-order central derivative dG/dt with G in G_0.
4. Peaks above 0.0007 are flickers and above 0.002 mechanical events. Events outside the molecular log G range are dropped as false positives.

The plots are an example trace with the events marked, its derivative with both thresholds, the share of each trace class, flickers per trace, and a before/after transition map. With holds you also get the conductance-time histogram with the most probable G.

Three choices to confirm. The SI doesn't pin these down, so GZero copies LabVIEW's defaults:

- ☐ **dt = 20 ms is LabVIEW's dt input**, so dG/dt = (G[i+1] - G[i-1]) / (2 x 0.02 s). The thresholds then correspond to steps of about 3 x 10^{-4} G_0 (flicker) and 9 x 10^{-4} G_0 (mechanical), since the dG/dt peak is about 2.2 times the step.
- ☐ **"Width 50" is the Peak Detector's quadratic-fit window**, not a minimum peak width. Read as a minimum width at half height, it rejects almost every step, because 200 Hz-filtered steps give derivative peaks only 20-30 samples wide at 10 kHz. Both readings are selectable.
- ☐ **Only rises are counted** (Peak Detector default), so a flicker up and back counts once. *Count = both* counts every step.

On synthetic steps a 4 x 10^{-4} G_0 flicker at 10^{-3} G_0 is found as a flicker, a jump from 10^{-6} to 1.6 x 10^{-3} G_0 as mechanical, and a flat trace with 3 % noise gives nothing.

EGaIn ensemble tab

Large-area junctions are analysed per junction, not per sweep. 20 junctions with 20 sweeps each are 20 independent measurements, not 400. The tab follows Reus et al. 2012, as XMe and GaussFit also do.

1. Each file is one junction. J = I / A, with A from the contact diameter (XMe's approach) or typed in.
2. A junction whose |J| reaches the short threshold (100 A/cm2) is a short. The yield is working junctions over all junctions.
3. Sweeps go onto a common bias grid. At each bias, log10|J| is histogrammed and fitted with a Gaussian to give the centre and $\sigma_{\log}$.

4. The confidence interval uses the number of junctions n_j as degrees of freedom (GaussFit's convention); formula below.
5. Also computed: log R = log|J(+V)| - log|J(-V)| per cycle, V_{trans} per cycle, and a single-level fit of the mean J(V) with free N.
6. Fit β across data sets with typed-in lengths d (formula below). Each point is weighted by the standard error of its mean (Simeone et al. 2013 report 0.92 per carbon).

$$\mathrm{CI} = t_{1-\alpha/2,\, n_j-1}\,\frac{\sigma_{\log}}{\sqrt{n_j-1}}, \qquad \log_{10}|J| = \log_{10} J_0 - \frac{\beta\, d}{\ln 10}$$

Check (simulated junctions)	True	Found
β, 5 sets of 30 junctions	0.9	0.904 +/- 0.016
Rectification log R at 0.5 V	0.30	0.30 +/- 0.08
Mean log|J| per bias	as simulated	inside its 95 % CI
β error coverage, 20 repeats	68 % within 1 σ	65 %

The first version weighted β by $\sigma_{\log}$ and claimed errors 3 times too small. Weighting by the standard error fixed that, which is what the coverage row shows.

Open-source code reviewed

Two public codes cover the same ground and were read in full for their EGaIn and I-V methods.

Code	Group	License	Modules	How GZero used it
XMe_DataAnalysis	Wenjing Hong, Xiamen University	Apache 2.0	Basic histograms, correlation, spectral clustering, PSD noise, I-V, single trace, thermopower, electrochemistry, EGaIn	EGaIn: J from contact diameter, log|J| per bias with Gaussian fits. The methods were re-implemented; XMe asks to be cited when compared with
GaussFit	Ryan Chiechi	GPL-3.0	EGaIn / CP-AFM parsing, Gaussian log|J|, rectification, V_{trans}, dJ/dV, lag filter, SLM	Junction-based CI, rectification and V_{trans} per cycle. Methods only: GPL code can't go into Apache-licensed GZero

GZero already covers the other Hong-group analyses in XMe: correlation, spectral clustering and flicker noise. Thermopower and electrochemical gating need measurement types GZero doesn't read yet.

Sources

Code: arman-rashid/GZero, release v1.4. The full method description is in the user manual (`docs/User_Manual.pdf`), sections 5.2, 6.4, 9.5, 9.8 and 10.2-10.5, references 26-35.

- M. Kamenetska et al., "Formation and evolution of single-molecule junctions", Phys. Rev. Lett. 102, 126803 (2009). doi:10.1103/PhysRevLett.102.126803

- L. A. Zotti et al., "Revealing the role of anchoring groups in the electrical conduction through single-molecule junctions", Small 6, 1529-1535 (2010). doi:10.1002/smll.200902227
- J. M. Beebe et al., "Transition from direct tunneling to field emission in metal-molecule-metal junctions", Phys. Rev. Lett. 97, 026801 (2006). doi:10.1103/PhysRevLett.97.026801
- I. Baldea, "Ambipolar transition voltage spectroscopy: analytical results and experimental agreement", Phys. Rev. B 85, 035442 (2012). doi:10.1103/PhysRevB.85.035442
- U. Rashid et al., "Mapping the extended ground state reactivity landscape of a photoswitchable molecule at a single molecular level", J. Am. Chem. Soc. 147, 830-840 (2025). doi:10.1021/jacs.4c13531 (SI sections 2.3-2.7)
- L. J. Reus et al., "Statistical tools for analyzing measurements of charge transport", J. Phys. Chem. C 116, 6714 (2012). doi:10.1021/jp210445y
- F. C. Simeone et al., "Defining the value of injection current and effective electrical contact area for EGaIn-based molecular tunneling junctions", J. Am. Chem. Soc. 135, 18131 (2013). doi:10.1021/ja408652h
- E. M. Opodi et al., "A single level tunneling model for molecular junctions: evaluating the simulation methods", Phys. Chem. Chem. Phys. 24, 11958 (2022). doi:10.1039/D1CP05807J
- XMe_DataAnalysis and GaussFit on GitHub